Year 7 Mathematics

Unit 4: Fractions

4.1 Working with Fractions

Working with Fractions

STARTER · 5 MIN

Date: _____ **Title: Working with Fractions**

LEARNING OBJECTIVES

- To be able to compare, simplify and work with fractions

STARTER *Answer in your book. No calculators.*

1. Find the HCF of 8 and 12

2. Which is greater, $\frac{1}{5}$ or $\frac{1}{3}$?

3. Which is greater, $\frac{5}{9}$ or $\frac{4}{9}$?

4. Find the HCF of 24 and 40

Starter — Answers

STARTER

Tick or correct in green pen

1. 4

2. $\frac{1}{3}$

3. $\frac{5}{9}$

4. 8

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can compare and order fractions	☐
SILVER	I can simplify fractions and write a part as a fraction of a whole	☐
GOLD	I can find a fraction of an amount	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Working with Fractions

INTRODUCTION · 3 MIN

Equivalent fractions have the same value: change the top and bottom in the same way.

Equivalent

Same value, different numbers.

$$\frac{1}{4} = \frac{2}{8} = \frac{3}{12}$$

Simplest form

Divide top and bottom by the HCF.

$$\frac{12}{20} = \frac{3}{5}$$ (HCF 4)

Compare

Use the same denominator.

$$\frac{3}{4} = \frac{15}{20} > \frac{14}{20} = \frac{7}{10}$$

Fraction of

÷ by the bottom, × by the top.

$\frac{3}{4}$ of $80 = 80 \div 4 \times 3 = 60$

Examples 1 and 2 — Equivalent Fractions and Simplest Form

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Hint Divide top and bottom by the HCF
Same change top and bottom

Example 1

Which fraction is equal to $\frac{4}{5}$

Choose from $\frac{8}{10}$, $\frac{3}{6}$ and $\frac{3}{4}$

Example 2

Write in its simplest form $\frac{12}{20}$

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

Multiply top and bottom by 2

$$\frac{4}{5} = \frac{4 \times 2}{5 \times 2}$$

$$= \frac{\mathbf{8}}{\mathbf{10}}$$

Tip: the value stays the same.

Example 2

Divide by the HCF of 12 and 20

HCF of 12 and 20 is 4

$$\frac{12}{20} = \frac{12 \div 4}{20 \div 4}$$

$$= \frac{\mathbf{3}}{\mathbf{5}}$$

Tip: stop when only 1 divides both.

Task 1

PRACTICE · 9 MIN

BRONZE *Simplify each fraction*

Hint Divide top and bottom by the HCF Same change top and bottom

1. $\frac{9}{12}$

2. $\frac{10}{15}$

3. $\frac{5}{10}$

4. $\frac{3}{9}$

5. $\frac{14}{35}$

★ $\frac{24}{40}$

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1 $\frac{3}{4}$

2 $\frac{2}{3}$

3 $\frac{1}{2}$

4 $\frac{1}{3}$

5 $\frac{2}{5}$

★ $\frac{3}{5}$

Examples 3 and 4 — Comparing Fractions

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Hint Use a common denominator Then compare the tops

Example 3

Which is greater?

$\frac{3}{4}$ or $\frac{7}{10}$

Example 4

Write < or > between

$\frac{3}{5} \square \frac{1}{2}$

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

Make the denominators the same: 20

$$\frac{3}{4} = \frac{15}{20}$$

$$\frac{7}{10} = \frac{14}{20}$$

So $\frac{3}{4} > \frac{7}{10}$

Tip: then compare the numerators.

Example 4

Use tenths

$$\frac{3}{5} = \frac{6}{10}$$

$$\frac{1}{2} = \frac{5}{10}$$

So $\frac{3}{5} > \frac{1}{2}$

Tip: > points to the smaller number.

Task 2

PRACTICE · 9 MIN

SILVER *Which is greater? Write < or >*

Hint Use a common denominator Then compare the tops

1. $\frac{1}{5} \square \frac{1}{3}$

2. $\frac{5}{9} \square \frac{4}{9}$

3. $\frac{5}{6} \square \frac{3}{4}$

4. $\frac{2}{7} \square \frac{3}{8}$

5. $\frac{3}{4} \square \frac{4}{5}$

★ Order, smallest first:

$\frac{5}{8}$, $\frac{1}{2}$, $\frac{2}{5}$

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1 $\frac{1}{5} < \frac{1}{3}$

2 $\frac{5}{9} > \frac{4}{9}$

3 $\frac{5}{6} > \frac{3}{4}$

4 $\frac{2}{7} < \frac{3}{8}$

5 $\frac{3}{4} < \frac{4}{5}$

★ $\frac{2}{5}, \frac{1}{2}, \frac{5}{8}$

Examples 5 and 6 — Fractions of Amounts and Parts of a Whole

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Hint ÷ bottom, then × top Fraction = part over whole

Example 5

(2 marks)

Work out $\frac{3}{5}$ of 35 m

Example 6

(2 marks)

A shop sells 18 rugby shirts.
9 are Harlequins shirts.

What fraction are Harlequins?
Give it in simplest form.

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

÷ by the bottom, × by the top

35 ÷ 5 = 7 m

7 × 3 = 21 m

Tip: find one fifth first.

Example 6

Part over whole, then simplify

$$\text{Fraction} = \frac{9}{18}$$

$$= \frac{\mathbf{1}}{\mathbf{2}}$$

Tip: label the part and the whole.

Task 3

PRACTICE · 9 MIN

GOLD *Exam-style: fractions of amounts*

Hint ÷ bottom, then × top Fraction = part over whole

1 $\frac{3}{4}$ of £80

2 $\frac{2}{3}$ of 150 mm

3 $\frac{4}{9}$ of 27 kg

4 5 of 15 clubs are British. What fraction is that?

5 Which is larger, $\frac{2}{3}$ of 18 m or $\frac{5}{6}$ of 24 m?

★ A 12 cm by 5 cm rectangle: $\frac{7}{10}$ is shaded. Area shaded?

Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

1. £60

2. 100 mm

3. 12 kg

4. $\frac{5}{15} = \frac{1}{3}$

5. 12 m and 20 m: 20 m

★ 60 ÷ 10 × 7 = 42 cm²

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Write $\frac{15}{20}$ in its simplest form.

SILVER Which is greater, $\frac{3}{4}$ or $\frac{4}{5}$?

GOLD Work out $\frac{2}{3}$ of 90.

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE $\frac{3}{4}$

SILVER $\frac{4}{5}$ $(\frac{16}{20} > \frac{15}{20})$

GOLD $90 \div 3 \times 2 \quad = 60$

Exam Practice — Simplifying Fractions

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

Hint Divide top and bottom by the HCF
Same change top and bottom

Question 1

(2 marks)

Write each fraction in its simplest form.

(a) $\frac{6}{10}$ (1)

.....

(b) $\frac{16}{28}$ (1)

.....

Question 2

(2 marks)

Find the missing numbers.

(a) $\frac{3}{4} = \frac{\square}{12}$ (1)

.....

(b) $\frac{2}{5} = \frac{8}{\square}$ (1)

.....

Exam Practice — Comparing Fractions

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint Use a common denominator
Then compare the tops

Question 3

(2 marks)

Which is larger, $\frac{3}{4}$ or $\frac{5}{6}$?

Show how you decide.

.....

Question 4

(2 marks)

Write these fractions in order of size.

Start with the smallest.

$\frac{2}{3}$ $\frac{1}{2}$ $\frac{7}{12}$ $\frac{3}{4}$

.....

Exam Practice — Fractions in Context

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint
÷ bottom, then × top
Fraction = part over whole

Question 5

(2 marks)

In a tennis match, Player B won 18 of the 27 games.

What fraction of the games did Player B win? Give it in simplest form.

.....

Question 6

(3 marks)

Amir spends $\frac{3}{8}$ of his \$48.

Lily spends $\frac{2}{5}$ of her \$45.

(a) How much does Amir spend? (1)

..... $

(b) Who spends more? (2)

.....

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

(a) $\frac{3}{5}$ — B1

(b) $\frac{4}{7}$ — B1

Tip: divide by the HCF.

Question 2

(a) 9 — B1

(b) 20 — B1

Tip: what was the bottom multiplied by?

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

$\frac{3}{4} = \frac{9}{12}$, $\frac{5}{6} = \frac{10}{12}$ M1

$\frac{5}{6}$ is larger A1

Tip: use a common denominator.

Question 4

In twelfths: 8, 6, 7, 9 M1

$\frac{1}{2}$, $\frac{7}{12}$, $\frac{2}{3}$, $\frac{3}{4}$ A1

Tip: change them all to twelfths.

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

Fraction $= \frac{18}{27}$ M1

$= \mathbf{\frac{2}{3}}$ A1

Tip: the HCF of 18 and 27 is 9.

Question 6

(a) 48 ÷ 8 × 3 = $18 B1

(b) 45 ÷ 5 × 2 = $18 M1

They spend the same A1

Tip: work out both before you decide.